

Lab 2 - Structure from Motion

How to use

1. Build

```
mkdir -p build
cd build
cmake -DCMAKE_BUILD_TYPE=Release ..
make -j$(nproc)
cd ..
```

2. Default run

```
./bin/basic_sfm datasets/test_data1.txt cloud1.ply
./bin/basic_sfm datasets/test_data2.txt cloud2.ply
```

Open the produced .ply in MeshLab. To clear the trackball sphere from the view, use *View > Show Trackball*.

3. Configuration knobs

Three environment variables can be prepended to the command to change the run-time configuration. They can be combined freely.

Variable	Default	Allowed values
SFM_LOSS	cauchy	cauchy huber null
SFM_SCALE	2.0	any positive double
SFM_SIDEWARD	0.5	0.0 to 1.0

4. Example commands

```
# Default (Cauchy at 2 * max_reproj_err_, sideward threshold 0.5)
./bin/basic_sfm datasets/test_data1.txt cloud1.ply

# Switch to Huber loss
SFM_LOSS=huber ./bin/basic_sfm datasets/test_data1.txt cloud1.ply

# Plain least squares (no robust loss)
SFM_LOSS=null ./bin/basic_sfm datasets/test_data1.txt cloud1.ply

# Tighter loss kernel scale
SFM_LOSS=cauchy SFM_SCALE=1.0 ./bin/basic_sfm datasets/test_data1.txt cloud1.ply

# Looser sideward threshold
SFM_SIDEWARD=0.3 ./bin/basic_sfm datasets/test_data1.txt cloud1.ply

# Multiple knobs combined
SFM_LOSS=huber SFM_SCALE=1.5 SFM_SIDEWARD=0.4 \
./bin/basic_sfm datasets/test_data1.txt cloud1.ply
```

5. Automatic sweep script

`./run_experiments.sh` runs three sweeps on both provided datasets, prints a table with point counts and registered cameras per configuration, and writes one `.ply` per cell into *results/* together with a per-cell log in *results/_logs/*.

```
chmod +x run_experiments.sh      # one time only
./run_experiments.sh
```

Sweeps run by the script: Experiment 1 compares Cauchy, Huber and NULL at scale 2.0; Experiment 2 sweeps the loss scale at 1.0, 1.5, 2.0, 3.0, 4.0 for both Cauchy and Huber; Experiment 3 sweeps the sideward threshold at 0.3, 0.4, 0.5, 0.6, 0.7. Total runtime is roughly 5 to 15 minutes on a recent laptop.